Ca' Foscari
University
of Venice

Performance Report of Vamsi Yerramsetti

March 15, 2022

Vamsi who is an Honors BSc Computer Science student from Radboud University had worked at my lab at Ca' Foscari University of Venice for the purpose of an academic internship facilitated by the Radboud Honors Academy of Science in the month of February 2022.

Prior to his arrival Vamsi read a paper, which explores the idea of abstracting the jigsaw puzzle problem as a consistent labeling problem with the help of relaxation labeling and modern Machine Learning/Artificial Intelligence concepts and techniques. After arrival, Vamsi started working on a python implementation of the strategy proposed in the paper. He worked mainly with PyTorch and experimented with various Machine Learning strategies. He would update his daily progress to his mentor Dr. Alessandro Torcinovich, who worked closely with Vamsi and taught him various AI and Computer Vision concepts during his visit.

Vamsi had expressed his interest in neural networks and the computational theory behind contextual information algorithms. We introduced him to the dynamics of such algorithms (relaxation labeling in depth) by making him read and understand two related scientific papers in this domain. He updated his mentor every day with his learnings and questions. In parallel, he was meeting with our Ph.D. student Marina Khoroshiltseva who taught him various in-depth concepts of Generative Adversarial Networks (GANs) and the machine learning frameworks behind them. Vamsi was asked to digest another research paper on GANs and was asked to implement a TensorFlow tutorial version of a GAN network on a trivial image dataset that generates a full image from a masked image. Vamsi executed all the above tasks with precision and his work was reviewed by his mentors. He additionally implemented a rotating function that provides the underlying foundation for solving major hurdles in Jigsaw puzzle solving. Apart from his research and assigned work, Vamsi showed interest in AI, ML, and pattern recognition and attended my lectures for the CS master's course "AI: Machine learning and Pattern Recognition".

Vamsi is passionate about neural networks and computer vision and we welcomed him to work and contribute to our projects in the future as he would surely be a valuable asset to the team.

He is highly motivated and has very good technical skills. He has also been participating actively in my lab activities and meetings, and his interaction with the other members of the group has always been very positive and proactive. Based on my personal experience, I'm therefore happy to support Vamsi's application in the strongest terms.

Dr.Alessandro's comments: Vamsi has been really committed to his tasks and worked tirelessly here while adapting to new situations. He should be proud of himself.

Dr. Prof. Marcello Pelillo

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